

[This question paper contains 16 printed pages]

Your Roll No. :

Sl. No. of Q. Paper : **7401** **J**

Unique Paper Code : 32341101 - OC

Name of the Course : **B.Sc.(Hons.) Computer Science**

Name of the Paper : Programming
Fundamentals Using C++

Semester : I

Time : 3 Hours **Maximum Marks : 75**

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
- (b) Question NO.1 is compulsory in **Section-A**.
- (c) Attempt any **four** questions from **Section-B**.
- (d) Parts of a question should be attempted together.

P.T.O.

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Section-A

1. (a) State error(s) if any, in each of the following code segments: 1×5=5

```
(i) int main()
    {
        const double p1;
        int n;
        p1 = 3.141;
        n = 2;
    }

(ii) class fun
    {   private:
        public:
            ~fun (int x );
    }

(iii) int myfun1(int x, y )
    {
        int myfun2(int x )
        {
            int t = x*2;
            return( t );
        }
        return y;
    }
```

- (iv) In the following code segment, assume that MyClass is the base class and YourClass is the derived class

```
MyClass ob1;
YourClass ob2;
ob2 = ob1;
```

- (v)

```
class A{ };
class B{ };
void myfunction() throw ( A )
{
    throw B();
}
```

- (b) Give the output that will be produced on execution of each of the following program segments: $2 \times 5 = 10$

- (i)

```
{
    string str1 = "Delhi ";
    string str2 = "Delhi1";
    int i = str1.compare
        (0,3,str2,0,3);
    cout << i << endl;
}
```

- (ii)

```
string str1 = "The c++
                standard";
int i = str1.find ("c");
int j = str1.rfind("c");
```

```

    cout << i << " " << j << endl;
(iii) int x = 3 , y = 4 , z = 5;
    if ( x < y++) || ( z++ > 0)
        cout << x << " " << y << " " << z;
(iv) int main()
    {
        for ( int i = 0 ; i < 8 ; i++)
        {
            if ( i%2 == 0)
                cout << i+1 << endl;
            else if (i%3 == 0) continue;
            else if (i%5 == 0) break;
            cout << " end of loop" << endl;
        }
        cout << "end of prog" << endl;
    }
(v) Assuming x1, x2, x3 are integer variables, give
    the output in each of the following two cases
    (consider each case independently) and justify your
    answer.
    if (x1)

```

```
if (x2)
if (x3)
x2 = 3;
else
x3 = 3;
cout << x1 << " " << x2 << "
" << x3;
```

Case 1 : if $x_1 = 0$, $x_2 = 1$ and $x_3 = 0$

Case 2 : if $x_1 = 1$, $x_2 = 0$ and $x_3 = 1$

(c) Give the output on execution of the following program segment:

4

```
int a = 5;
int b = 8;
int c;
int* p1 = &b;
int* p2;
int * p3;
p2 = p1;
p3 = &c;
```

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```
p1 = &a;
*p2 = 8;
*p3 = *p1;
*p3 = a + *p2 + *(&c);
cout << *p1 << " " << *p2 << " "
<< *p3 << *p1 + *p2 << endl;
```

(d) What for do we use key word `const` at the end of a function header? 2

(e) What will be the value of `z` on execution of the following statement if: 2

`x = 4 , y = 4:`

`z = (x < y ? -2 : (x == y ? 0 : 2));`

(f) Write C++ statements to illustrate how input output stream flags can be used to produce the output that is :

4

(i) left justified

(ii) right justified

- (g) If the value of $x = 3$ and $y = 5$, what will be the values of x and y on evaluation of each of the following expressions independently? 4

(i) $x++ + y$

(ii) $++x + 4$

(iii) $-x + y++$

(iv) $--x+--y$

- (h) if $x = 5$, $y = 0$, and $z = -2$, what will be the value of the following expressions: 4

(i) $x \ \&\& \ y \ || \ z$

(ii) $(x \ || \ y) \ \&\& \ z$

Section - B

2. (a) In the following code, point out the statements in which copy constructor will be called. 4

```
class Rational
{
    private :
    int num,den;
    public :
```

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```
Rational() { }

Rational (int n , int d): num(n),den
(d)

{ }

Rational(const Rational& r):
num(r.num),
den(r.den)
{
    cout << "COPY CONSTRUCTOR
CALLED\n";
}

};

Rational f(Rational r )
{
    Rational s = r;
    return s;
}

void main()
{
```



```
Rational x(22, 7);
```

```
Rational y(x);
```

```
x = f(y) ;
```

```
}
```

- (b) What are abstract classes ? Illustrate the use of abstract classes with an example. 6

3. (a) How is function overloading different from function overriding ? Give an example of each. 4

- (b) Write a program to read an integer n , followed by n integers. Compute the average of these numbers. The program should throw an exception for the following :

6

- (i) n is zero
- (ii) n is negative

The program should be able to handle any other exception thrown.

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4. (a) Write a function `checkSorted` that accepts two parameters 6

(i) Size of an integer array

(ii) An array of integers

The function `checkSorted` should return true if input array is sorted in descending order and false otherwise.

- (b) What will be the output on execution of the following program; 4

```
class Compute
{
    virtual int add(int a , int b )
    {
        return a + b ;
    }
    int add(int a , int b , int c )
    {
        return a + b + c ;
    }
};

Class Simple: public Compute
{
```

```
virtual int add (int a, int b)
{
    return a + b + 2 ;
}

};

void main( )
{
    Compute ob1;
    Compute * p;
    Simple ob3;
    p = &ob3;
    cout << p-> add(4,5 ) ;
    p = & ob1;
    cout << p-> add (6, 7 );
    cout << ob1.add (10,20);
    cout << ob1.add (10, 20, 30);
    return;
}
```

5. (a) Define a class fraction having data members numerator and denominator. Write member functions to do the following: 6

(i) create an object fr1 for the fraction 5/7

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(ii) increment the fraction using postfix operator ++
(using operator overloading),

`fr++` should update `fr` to `12/7`, by modifying
the value of the numerator .

(b) Write a C++ program that reads a text file
`f1.txt` and creates another file `f2.txt` so
that every sequence of consecutive blank
spaces in `f1.txt` is replaced by a single
character '@' in the output file `f2.txt`.

4

6. (a) If `str1` and `str3` are C++ strings, write C++
statements for the following:

4

- (i) to obtain the substring `str3` comprising the first
four characters of `str1`
- (ii) to find the position of the first occurrence of `str3`
in the string `str1`

(iii) to find the second occurrence of the string `str3` in `str1`

(b) Write a program to create a class called `Person` having the following data members: (i) `Aadhar No`, `Name`, `birth_year`. Write a following function to: 6

(i) create an object for a person using parameterized constructor.

(ii) `calculate_age` which accepts as input parameter the current year and calculates the person's age in years.

(iii) `display` to print all the details for a given object of the class along with the calculated age.

7. (a) class demo

```
{  
    public:  
        static int objectcount;
```

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```
int a;
demo(int k)
{
    a = k;
    cout << ++objectcount << a;
}

};

int demo :: objectcount = 0;
void main( )
{
    demo ob1( 4 ), ob2( 5 ), ob2( 6 );
    cout << "n_objects created"
    << demo::objectcount;
}
```

6

(b) Give the output on the execution of the following code:

4

```
class myClass
{
```

```
int x , y;
public :
myClass(int x1, int y1)
{
    x = x1;
    y = y1;
}
friend int sum (myClass ob1)
{
    int z ;
    z = ob1.x + ob1.y ;
    return z;
}
void increment( )
{
    x ++;
    y ++;
    cout << x << y;
}
};

void main()
{
    int y1 ;
```

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```
myClass ob1( 4, 5 );  
y1= sum (ob1);  
cout << y1;  
ob1.increment();  
}
```